

Homework 05 Functions and Permutation Tests

Due by 11:59pm, Tuesday 10.14.25

S&DS 2300/5300/ENV 757

Submission Details

Questions 1 and 2 use data from both 2017 and 2018 New Haven Road Races - in particular, we look at 5k run times. You can get data for 2018 HERE (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/NHRR2018.csv) and for 2017 HERE (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/NHRR2017.csv).

1) Function for Data Cleaning (20 points)

(1.1) (5 pts) Load in both .csv files into objects called `nh2017` and `nh2018`. Use `head()`, `names()`, and `str()` to check if both datasets have the same variable names and the same format (i.e does each variable have the same format in each dataset). Comment on what you observe.

```
nh2017 <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/NHRR2017.csv")
nh2018 <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/NHRR2018.csv")
```

```
head(nh2017)
```

##	No.	Name	City	Div	Time	Pace	Nettime
## 1	3376	Patrick Dooley	Brooklyn	M30-39	15:17	4:56	15:16
## 2	2884	Calvin Park	Trumbull	M20-29	15:19	4:56	15:18
## 3	2839	Jake Duckworth	Monroe	M20-29	15:29	4:59	15:28
## 4	1150	Scott Rodilitz	New Haven	M20-29	15:37	5:02	15:36
## 5	1567	Robert Dillon	Shelton	M13-19	15:47	5:05	15:46
## 6	4256	Nicholas Migani	Higganum	M20-29	16:00	5:09	15:59

```
names(nh2017)
```

```
## [1] "No." "Name" "City" "Div" "Time" "Pace" "Nettime"
```

```
str(nh2017)
```

```
## 'data.frame': 2736 obs. of 7 variables:
## $ No. : int 3376 2884 2839 1150 1567 4256 3963 4307 5131 5740 ...
## $ Name : chr "Patrick Dooley" "Calvin Park" "Jake Duckworth" "Scott Rodilitz" ...
## $ City : chr "Brooklyn" "Trumbull" "Monroe" "New Haven" ...
## $ Div : chr "M30-39" "M20-29" "M20-29" "M20-29" ...
## $ Time : chr "15:17" "15:19" "15:29" "15:37" ...
## $ Pace : chr "4:56" "4:56" "4:59" "5:02" ...
## $ Nettime: chr "15:16" "15:18" "15:28" "15:36" ...
```

```
head(nh2018)
```

```
##      No.      Name      City      Div      Time      Pace      Nettime
## 1 4606 Matthew Farrell Glastonbury M13-19 15:19 4:56    15:19
## 2 2643   Robert Dillon   Shelton M13-19 15:38 5:02    15:38
## 3 4037   Azaan Dawson   New Haven M13-19 15:51 5:07    15:51
## 4 3712   Travis Martin   New Haven M13-19 16:03 5:10    16:00
## 5 4633   Mustafe Dahir Wallingford M13-19 16:19 5:15    16:17
## 6 2731     Ethan Puc    Naugatuck M13-19 16:27 5:18    16:25
```

```
names(nh2018)
```

```
## [1] "No."      "Name"     "City"     "Div"      "Time"     "Pace"     "Nettime"
```

```
str(nh2018)
```

```
## 'data.frame':    2685 obs. of  7 variables:
## $ No.      : int  4606 2643 4037 3712 4633 2731 4800 3710 4618 3142 ...
## $ Name     : chr   "Matthew Farrell" "Robert Dillon" "Azaan Dawson" "Travis Martin" ...
## $ City     : chr   "Glastonbury" "Shelton" "New Haven" "New Haven" ...
## $ Div      : chr   "M13-19" "M13-19" "M13-19" "M13-19" ...
## $ Time     : chr   "15:19" "15:38" "15:51" "16:03" ...
## $ Pace     : chr   "4:56" "5:02" "5:07" "5:10" ...
## $ Nettime  : chr   "15:19" "15:38" "15:51" "16:00" ...
```

Based on the `names()` and `str()`, it looks like both `nh2017` and `nh2018` have the same variable names and the same format (all the variable names have the same structure in both versions), and there's no striking difference in how the data is entered.

(1.2) (15 pts) Since the two datasets seem to have the same structure, we can write a function that creates new variables in each dataset. This function will be called `cleanNHData()`. As a first step, I've already included code to load the `lubridate` package and define a function called `convertTimes()` similar to that we used in Class 10.

I've started the outline of the function below. Your job is to follow the exact process we used in class 9 to clean the 2018 data. You need to replace each comment line in the `cleanNHData()` function with the code that will perform this task. You literally just need to find the relevant line in the class code and put this into the `cleanNHData()` function. The one exception is a new line you'll need to write that deletes rows where `Name` is missing (i.e. equal to `""`)

Then, run the function on `nh2017` and `nh2018` to replace each of these datasets with the cleaned up version of themselves.

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
##
##      date, intersect, setdiff, union
```

```

convertTimes <- function(v) {
  hourplus <- nchar(v) == 7
  wrongformat <- nchar(v) == 8
  outtimes <- ms(v)
  if (sum(hourplus) > 0) { # if there is at least 1 time that exceeds 1 hr
    outtimes[hourplus] <- hms(v[hourplus])
  }
  if (sum(wrongformat) > 0) { # if there is at least 1 time in wrong format
    outtimes[wrongformat] <- ms(substr(v[wrongformat],1,5))
  }
  outtimes <- as.numeric(outtimes)/60
  return(outtimes)
}

```

```

cleanNHData <- function(data) {
  #Replace Div = "" with NA
  data$Div[data$Div == ""] <- NA
  #Make a dataset variable called Gender from the variable Div
  data$Gender <- substr(data$Div, 1, 1)
  #Make a dataset variable called AgeGrp from the variable Div
  data$AgeGrp <- substr(data$Div, 2, nchar(data$Div))
  #Make a dataset variable called Nettime_min using the convertTimes function
  data$Nettime_min <- convertTimes(data$Nettime)
  #Make a dataset variable called Time_min using the convertTimes function
  data$Time_min <- convertTimes(data$Time)
  #Make a dataset variable called Pace_min using the convertTimes function
  data$Pace_min <- convertTimes(data$Pace)

  #Replace dataset with same dataset such that Name is not equal to ""
  data <- data[data$Name != "", ]

  #Return the dataset
  return(data)
}

```

```

#run cleanNHData on nh2018 and nh2017 and replace these with the cleaned up versions of themselves
nh2018 <- cleanNHData(nh2018)

```

```

## Warning in .parse_hms(..., order = "MS", quiet = quiet): Some strings failed to
## parse
## Warning in .parse_hms(..., order = "MS", quiet = quiet): Some strings failed to
## parse
## Warning in .parse_hms(..., order = "MS", quiet = quiet): Some strings failed to
## parse

```

```

nh2017 <- cleanNHData(nh2017)

```

```

## Warning in .parse_hms(..., order = "MS", quiet = quiet): Some strings failed to
## parse
## Warning in .parse_hms(..., order = "MS", quiet = quiet): Some strings failed to
## parse

```

2) Repeat Runners Dataset (35 points)

We now create a dataset that looks at times of runners who ran in both 2018 and 2017.

(2.1) (5 pts) We'll have problems if we have instances of two runners having the same name. A crude fix is to delete the second occurrence of anyone with a duplicate name.

Run the code below to see how the function `duplicated()` works:

```
duplicated(c("cat", "cat", "dog", "llama"))
```

```
## [1] FALSE  TRUE FALSE FALSE
```

Essentially, this returns a vector that is `FALSE` if an observation value is the first occurrence of this value and `TRUE` when a value has been seen before.

To merge our two datasets, we need to start with unique `Name` values in each dataset. Using the `duplicated()` function, create two new dataframes called `nh2018Unq` and `nh2017Unq` so that each only retains observations for the first occurrence of each value of `Name` (if you use the `!` operator, this is two short lines of code).

Get the dimensions of each of the four relevant dataframes. How many observations were eliminated from each year?

```
nh2018Unq <- nh2018[!duplicated(nh2018$Name), ]  
nh2017Unq <- nh2017[!duplicated(nh2017$Name), ]
```

```
dim(nh2018)
```

```
## [1] 2685  12
```

```
dim(nh2018Unq)
```

```
## [1] 2640  12
```

```
dim(nh2017)
```

```
## [1] 2727  12
```

```
dim(nh2017Unq)
```

```
## [1] 2720  12
```

For 2018, we can see that 45 observations were eliminated. For 2017, we can see that 7 observations were eliminated.

(2.2) (5 pts) Next, we need to get a list of names that occur in both datasets. Run the code below to see how the `intersect()` function works.

```
intersect(c("cat", "dog", "llama"), c("cat", "llama", "chincilla"))
```

```
## [1] "cat"  "llama"
```

Using the `intersect()` function, create an object called `repeatrunners` that is a list of names of people who ran in both years. How many runners ran in both years?

```
repeatrunners <- intersect(nh2018Unq$Name, nh2017Unq$Name)
length(repeatrunners)
```

```
## [1] 986
```

986 runners ran in both years.

(2.3) (15 pts) The code below will create a combined dataset called `nhcombined`. Your job in this section is to write a one or two line comment above each line of code to describe what the line does. You'll want to run each line, probably see what the result was, and in some cases use the help file for some functions to see what the function does (i.e. for the `merge()` function). Make sure you remove `eval = FALSE` in the `r` chunk.

```
# Make a vector of names 'w' that contain the names in nh2018Unq that are also in repeat runners
w <- nh2018Unq$Name %in% repeatrunners

# Make a new dataframe that has Name, Gender, and Nettime_min from the 'w' participants in nh2018
Unq
nhcombined <- data.frame(Name = nh2018Unq$Name[w],
                        Gender = nh2018Unq$Gender[w],
                        Nettime_2018 = nh2018Unq$Nettime_min[w])

# Merge nhcombined with nh2017, only keeping "Name" and "Nettime_min" to add each runner's 2017 t
ime
nhcombined <- merge(nhcombined, nh2017Unq[, c("Name", "Nettime_min")])

# Removed any runner from nhcombined that has missing Gender
nhcombined <- nhcombined[!is.na(nhcombined$Gender),]

# Renames the fourth column of nhcombined to "Nettime_2017" to reflect that it's the 2017 time fo
r the runner
colnames(nhcombined)[4] <- "Nettime_2017"

# Show the dimensions of nhcombined
dim(nhcombined)
```

```
## [1] 985    4
```

```
# Show the first 6 rows of nhcombined
head(nhcombined)
```

```
##           Name Gender Nettime_2018 Nettime_2017
## 1   Abbey Shaw     F    39.25000    40.25000
## 2   Abby Dziura     F    39.03333    35.63333
## 3   Abby Ganun     F    40.08333    44.65000
## 4   Abi Hawkins     F    35.86667    27.56667
## 5  Abigail Murphy     F    32.88333    34.06667
## 6 Abraham Cordero     M    29.63333    31.83333
```

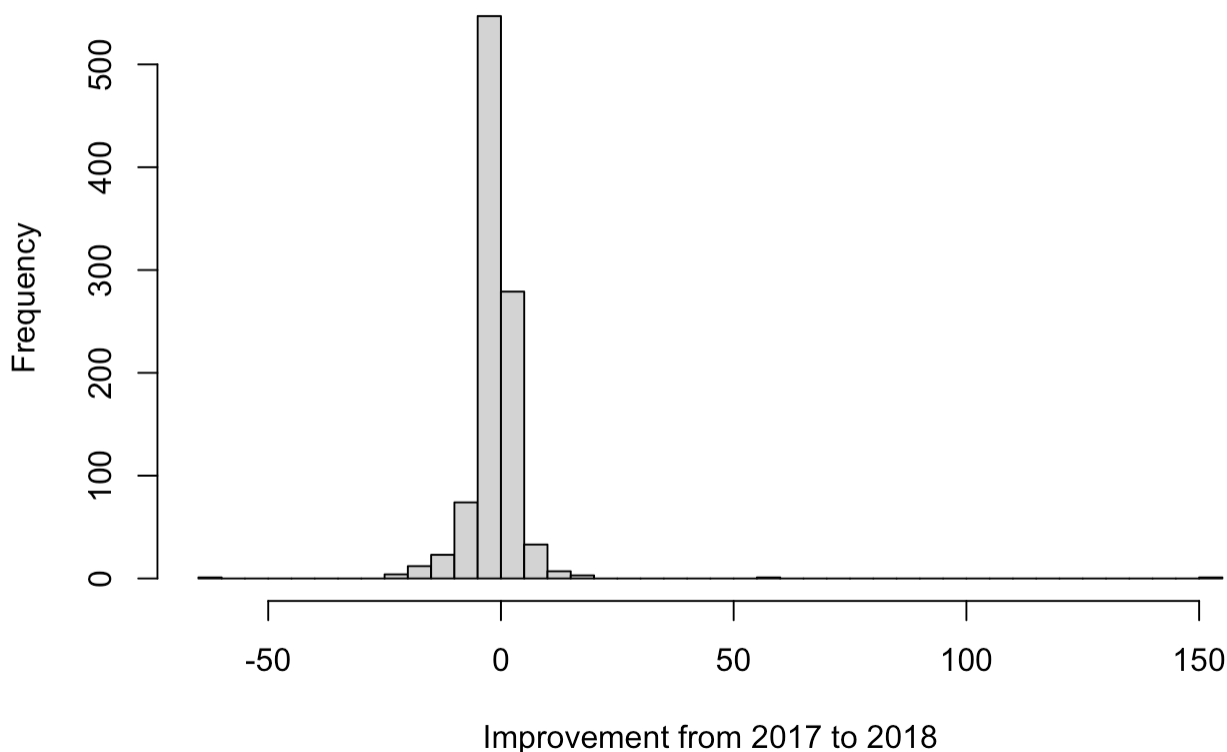
(2.4) (5 pts) Create a new variable in the data frame `nhcombined` called `improvement` that is the improvement in run time from 2017 to 2018 (a positive number here should indicate an improvement, a negative number means they did worse in 2018). Get summary statistics for `nhcombined`. Then make a histogram of `improvement`. Comment on the summary statistics and what you observe in the histogram.

```
nhcombined$improvement <- nhcombined$Nettime_2017- nhcombined$Nettime_2018
summary(nhcombined)
```

```
##      Name      Gender      Nettime_2018      Nettime_2017
## Length:985      Length:985      Min.   : 15.63      Min.   : 15.30
## Class :character Class :character 1st Qu.: 26.12      1st Qu.: 25.43
## Mode  :character Mode  :character Median  : 30.60      Median : 29.37
##                                     Mean   : 32.04      Mean   : 30.93
##                                     3rd Qu.: 36.28      3rd Qu.: 34.32
##                                     Max.   :132.28      Max.   :188.08
## improvement
## Min.   :-64.5167
## 1st Qu.: -2.6000
## Median : -0.9333
## Mean    : -1.1156
## 3rd Qu.:  0.5333
## Max.    :150.2667
```

```
hist(nhcombined$improvement,
     breaks = 50,
     xlab = "Improvement from 2017 to 2018",
     ylab = "Frequency",
     main = "Histogram of Time Improvement from 2017 to 2018")
```

Histogram of Time Improvement from 2017 to 2018



Based on the summary statistics and the histogram, we see that the improvement value tends to lean negative, meaning that people were generally slower in the 2018 race compared to the 2017 race.

(2.5) (5 pts) You'll notice a few extreme values (i.e. people got amazingly better or worse). Print the rows of `nhcombined` that

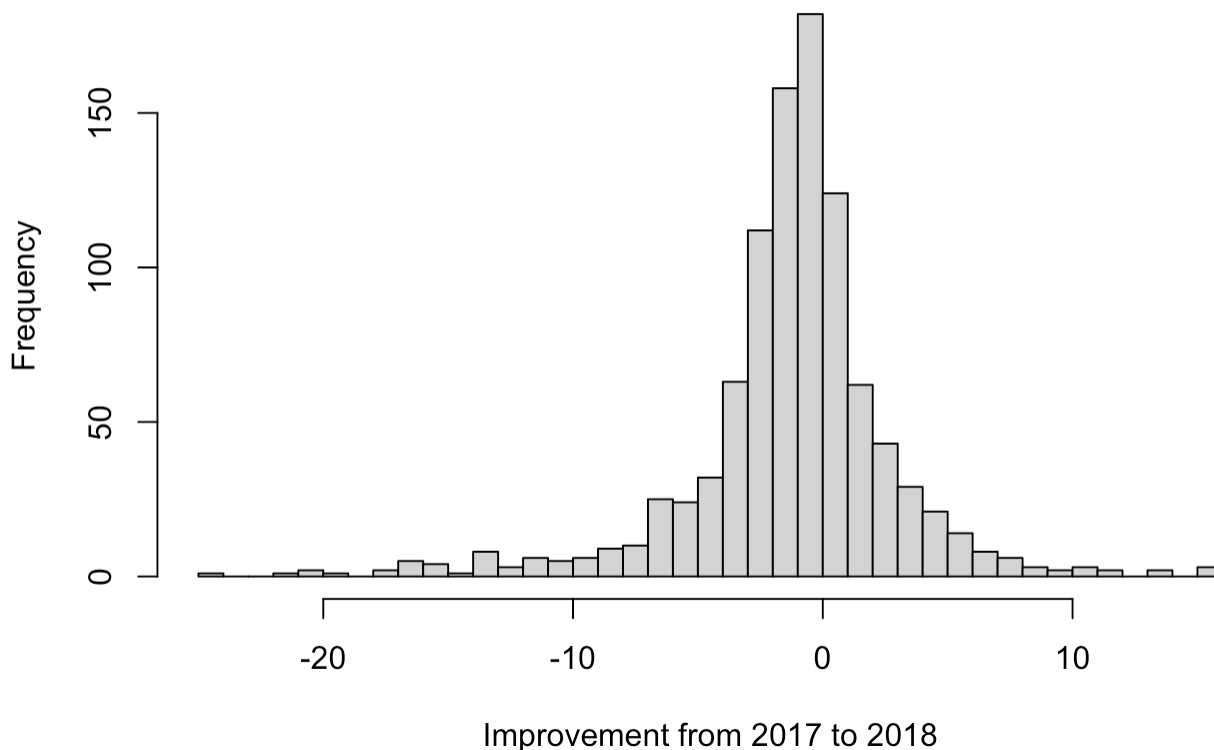
had improvement times of more than 50 in absolute value. Update the `nhcombined` dataframe to exclude these rows and make the histogram again.

```
nhcombined_extreme <- nhcombined[abs(nhcombined$improvement) > 50, ]
nhcombined_extreme
```

```
##           Name Gender Nettime_2018 Nettime_2017 improvement
## 483 Julius Bloom      M      30.28333      87.41667      57.13333
## 594 Lina Alpert       F     109.51667      45.00000     -64.51667
## 706 Mike Trumbley     M      37.81667     188.08333     150.26667
```

```
nhcombined <- nhcombined[abs(nhcombined$improvement) < 50, ]
hist(nhcombined$improvement,
     breaks = 50,
     xlab = "Improvement from 2017 to 2018",
     ylab = "Frequency",
     main = "Histogram of Time Improvement from 2017 to 2018")
```

Histogram of Time Improvement from 2017 to 2018



3) Changes in Measles Vaccination Rates in the Past 8 Years (45 pts)

Question 3 uses data from the 2016 and 2024 World Bank datasets. You can get data for 2016 [HERE](https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2016.csv) (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2016.csv) and for 2024 [HERE](https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2024.csv) (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2024.csv).

(3.1) (10 pts) Read in the datasets. Get and show the names on each dataset. Confirm that the dimensions are the same between datasets. Then, modify each dataset so that it only contains “Country”, “Measles”, and “GNI”.

Following the example in question 2, combine these datasets together. You’ll need to rename “Measles” and “GNI” in each dataset before combining based on Country (something like “Measles_24”, “Measles_16”, etc.) In the combined dataset,

remove any observations that are missing for either Measles variable or for GNI in 2024.

Create a new factor variable that identifies countries as having GNI in 2024 greater than 8000 or less than or equal to 8000.

Finally, calculate a variable that is the change in Measles vaccination rates (2024 minus 2016) per country.

Show the first 10 rows of the final dataset.

```
wb_16 <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2016.csv")
wb_24 <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2024.csv")

names(wb_16)
```

##	[1]	"Country"	"Code"	"Population"	"Rural"
##	[5]	"GNI"	"IncomeTop10"	"Imports"	"Exports"
##	[9]	"Military"	"Cell"	"Fertility66"	"Fertility16"
##	[13]	"Measles"	"InfMort"	"LifeExp"	"PM2.5"
##	[17]	"Diesel"	"C02"	"EnergyUse"	"FossilPct"
##	[21]	"Forest94"	"Forest14"	"Deforestation"	"GunTotal"
##	[25]	"GunHomicide"	"GunSuicide"	"GunUnint"	"GunUndet"
##	[29]	"GunsPer100"			

```
names(wb_24)
```

##	[1]	"Country"	"Population"	"Rural"	"GNI"	"Imports"
##	[6]	"Exports"	"Military"	"Cell"	"Fertility"	"Measles"
##	[11]	"InfMort"	"LifeExp"	"PM2.5"	"C02"	"EnergyUse"
##	[16]	"Renewable"	"Debt"	"EU"		

```
dim(wb_16)
```

```
## [1] 217 29
```

```
dim(wb_24)
```

```
## [1] 217 18
```



```

wb_16 <- wb_16[, c("Country", "Measles", "GNI")]
wb_24 <- wb_24[, c("Country", "Measles", "GNI")]

wb_combined <- data.frame(Country = wb_16$Country,
                          Measles_16 = wb_16$Measles,
                          GNI_16 = wb_16$GNI)

wb_combined <- merge(wb_combined, wb_24[, c("Country", "Measles", "GNI")])
colnames(wb_combined)[4] <- "Measles_24"
colnames(wb_combined)[5] <- "GNI_24"

wb_combined <- wb_combined[!is.na(wb_combined$Measles_16)
                           & !is.na(wb_combined$Measles_24)
                           & !is.na(wb_combined$GNI_24), ]

wb_combined$GNI_8000 <- factor(ifelse(wb_combined$GNI_24 > 8000, "Higher", "Lower"))
wb_combined$Measles_diff <- wb_combined$Measles_24 - wb_combined$Measles_16

head(wb_combined, 10)

```

```

##          Country Measles_16 GNI_16 Measles_24 GNI_24 GNI_8000
## 1      Afghanistan      62   580      68    360      Lower
## 2          Albania      96  4320      86   6770      Lower
## 3          Algeria      94  4360      79   4490      Lower
## 5          Andorra      97    NA      98  50080      Higher
## 6          Angola      49  3450      37   1870      Lower
## 7  Antigua and Barbuda      98 13560      99  18710      Higher
## 8          Argentina      90 11940      83  11590      Higher
## 9          Armenia      97   3770      95   5960      Lower
## 11         Australia      95 54130      96  60820      Higher
## 12         Austria      95 45850      95  55720      Higher
##  Measles_diff
## 1           6
## 2        -10
## 3        -15
## 5           1
## 6        -12
## 7           1
## 8          -7
## 9          -2
## 11          1
## 12          0

```

(3.2) (10 pts) Calculate and display summary statistics for the change in Measles vaccination rates overall. Make a histogram of these changes and add a vertical line at the value which indicates no change. Discuss what you observe in a few sentences.

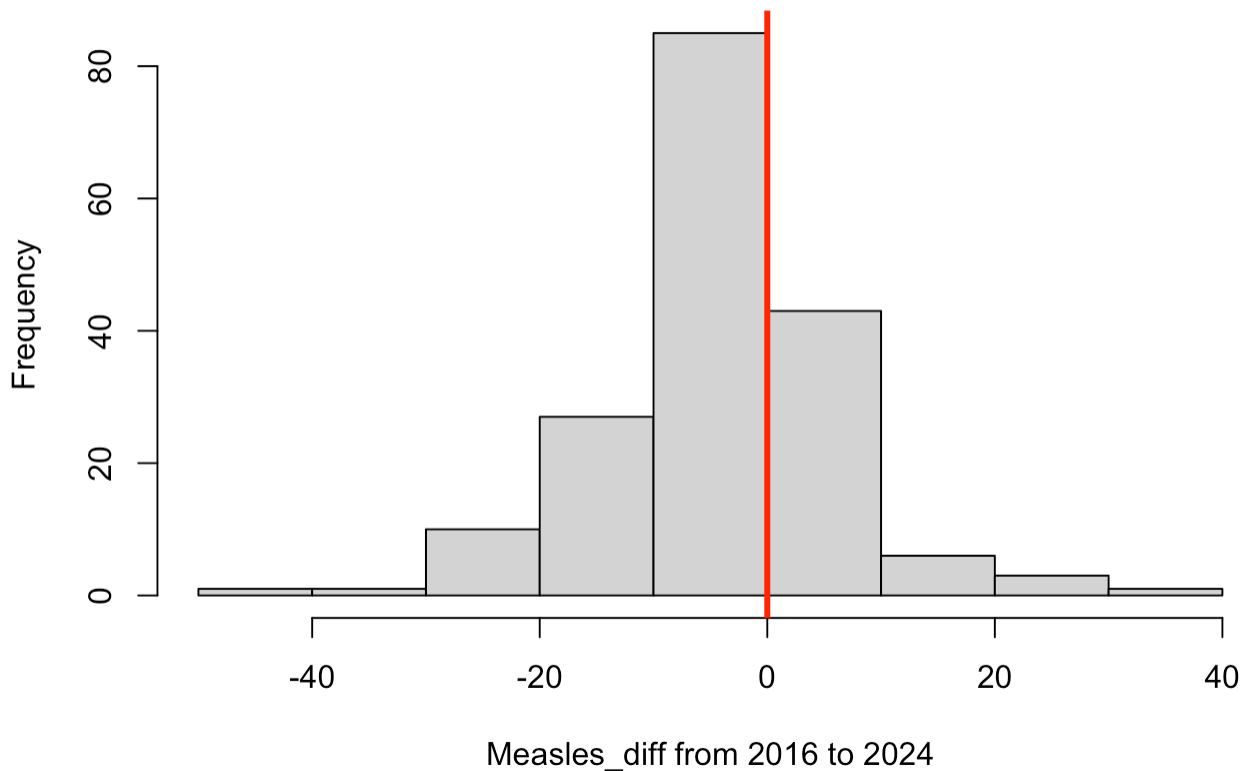
Make a side-by-side boxplot to see differences between the change in Measles vaccination rates between Countries with GNI < 8000 and countries with GNI > 8000. Does there appear to be any difference between groups? Comment both on center and spread.

```
summary(wb_combined$Measles_diff)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	-45.000	-7.000	-1.000	-3.153	1.000	32.000

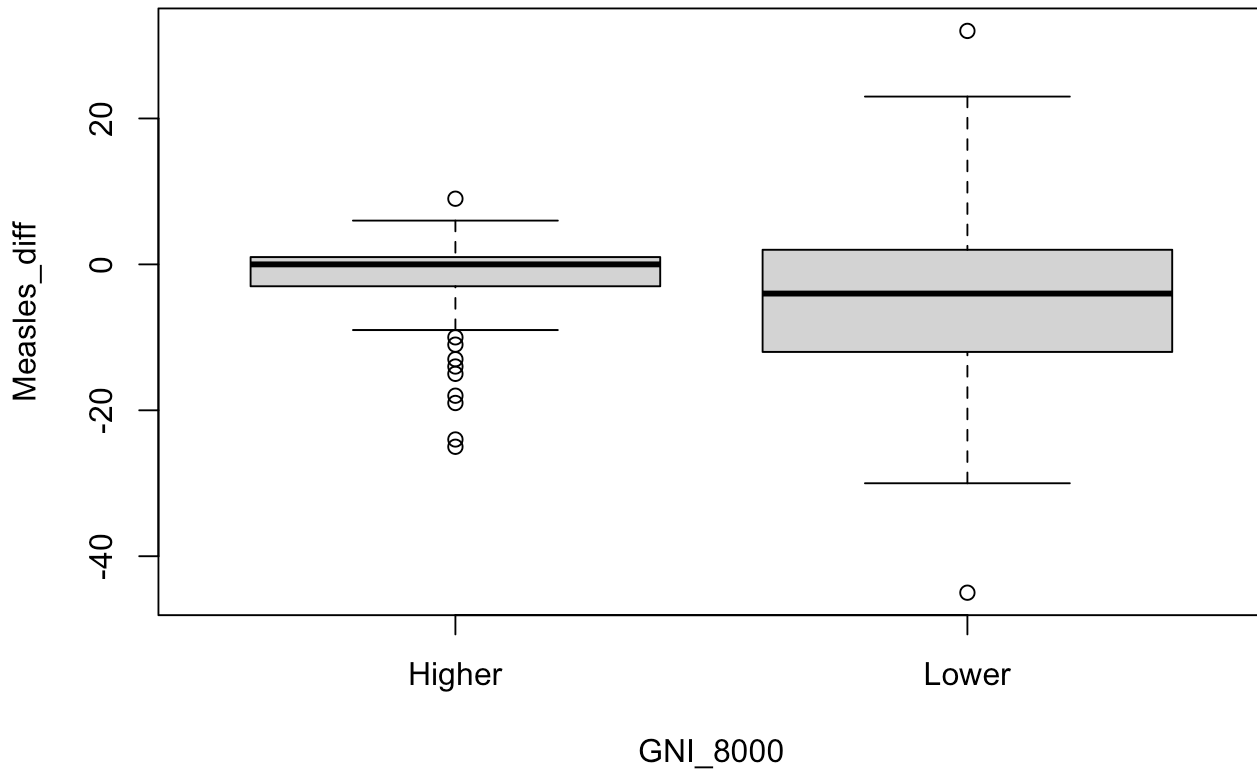
```
hist(wb_combined$Measles_diff,  
      xlab = "Measles_diff from 2016 to 2024",  
      ylab = "Frequency",  
      main = "Histogram of Measles_diff from 2016 to 2024")  
abline(v = 0, col = "red", lwd = 3)
```

Histogram of Measles_diff from 2016 to 2024



```
boxplot(Measles_diff ~ GNI_8000, data = wb_combined,  
        main = "Box plot of Measles_diff by GNI_8000")
```

Box plot of Measles_diff by GNI_8000



According to the summary statistics and the histogram, we can see that measles vaccination actually went down in 2024 compared to 2016, but that most of it seems to be a small change (not extreme). For the boxplot, it seems that countries with a GNI above 8000 tend to center around a Measles_diff of 0 while countries with a GNI below 8000 tend to center around a Measles_diff of a little below 0. In terms of spread, countries with a GNI above 8000 seem to be a lot tighter while countries with a GNI below 8000 seem to be a lot more spread out.

(3.3) (10 pts) Create a 95% bootstrap confidence interval for the mean change in measles vaccination rates among countries with GNI < 8000. Do the same for countries with a GNI > 8000. You don't need to make any histograms of your bootstrap results, and you don't need to use the `t.test()` function. You also are not comparing the means of these two groups - you're getting separate intervals for each GNI group. Display the intervals and discuss what you observe.

```
# To make grading easier, please leave the following line of code in your assignment
set.seed(230)
```

```
# FILL IN REMAINING CODE
```

```
N <- 10000
measles_diff_high <- rep(NA, N)
measles_diff_low <- rep(NA, N)
for (i in 1:N){
  s_diff_high <- sample(wb_combined$Measles_diff[wb_combined$GNI_8000 == "Higher"],
                        sum(wb_combined$GNI_8000 == "Higher"),
                        replace= TRUE)
  s_diff_low <- sample(wb_combined$Measles_diff[wb_combined$GNI_8000 == "Lower"],
                      sum(wb_combined$GNI_8000 == "Lower"),
                      replace= TRUE)
  measles_diff_high[i] <- mean(s_diff_high)
  measles_diff_low[i] <- mean(s_diff_low)
}

ci_high <- quantile(measles_diff_high, c(0.025, 0.975))
round(ci_high, 2)
```

```
## 2.5% 97.5%
## -3.59 -0.75
```

```
ci_low <- quantile(measles_diff_low, c(0.025, 0.975))
round(ci_low, 2)
```

```
## 2.5% 97.5%
## -6.24 -1.64
```

In our confidence intervals, we see that when $GNI > 8000$, our mean falls somewhere within -3.59 and -0.75. When $GNI \leq 8000$, we see that our mean falls somewhere within -6.24 and -1.64, both values lower than when $GNI > 8000$, and its range is also larger.

(3.4) (15 pts) Using a permutation test, examine whether there a significant difference in the **MEDIAN** change in vaccination rates between high GNI countries and low GNI countries (calculate as high - low). Use a significance level of 0.01. Be sure to state (in words is fine) the null and alternative hypotheses, and discuss your conclusion. Be sure to include a histogram of results and add a vertical line that shows that observed difference in medians (see example in code from class).

```
# To make grading easier, please leave the following line of code in your assignment
set.seed(230)
```

```
# FILL IN REMAINING CODE
```

```
actualdiff <- median(wb_combined$Measles_diff[wb_combined$GNI_8000 == "Higher"]) -
  median(wb_combined$Measles_diff[wb_combined$GNI_8000 == "Lower"])
```

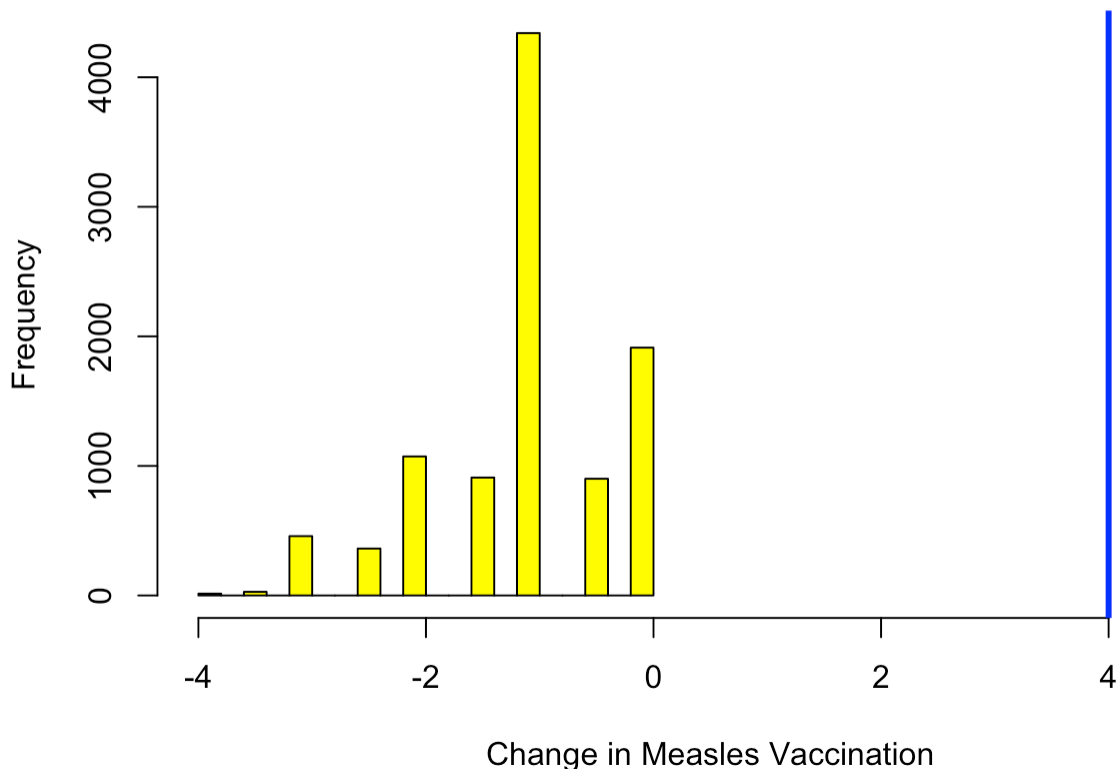
```
N <- 10000
```

```
diffvals <- rep(NA, N)
```

```
for (i in 1:N){
  fake_GNI <- sample(wb_combined$GNI_8000)
  diffvals[i] <- median(wb_combined$Measles_diff[fake_GNI == "Higher"])
    - median(wb_combined$Measles_diff[fake_GNI == "Lower"])
}
```

```
hist(diffvals, col = "yellow", main = "Permuted Sample Median Diff in Measles Vaccination Change",
      xlab = "Change in Measles Vaccination", xlim = c(-4, 5))
abline(v = actualdiff, col = "blue", lwd = 3)
```

Permuted Sample Median Diff in Measles Vaccination Change



```
mean(abs(diffvals) >= abs(actualdiff))
```

```
## [1] 0.0014
```

Null hypothesis: There is no difference in the median change in measles vaccination between countries with high GNI and countries with low GNI Alternative hypothesis: There is a difference in the median change in measles vaccination between countries with high GNI and countries with low GNI Because we are using a significance level of 0.01 and our p-value is

0.0014, we reject the null hypothesis

4. Pseudo-coding

A public health group in New Haven is studying patterns of childhood and respiratory illnesses over the past 25 years, focusing on the fall season from August through mid-December. The dataset includes weekly counts of hand, foot, and mouth disease and strep throat from the first full week of August through the last full week of December. There is also information on the academic and break/holiday schedules for Greater New Haven colleges as well as the public school system. Vaccination rates for each year (not week) are also available. Write pseudocode describing how you would explore how yearly illness incidence changes over time, test whether rates differ before and after school starts, compare high- versus low-vaccination years, and assess whether the observed differences are statistically significant.

- *Read in data using read.csv()*
- *Use names(), dim(), and str() to get a general idea of the data*
- *Explore how yearly illness incidence changes over time*
 - *Make a plot of yearly illness incidence*
 - *x axis will be time in years*
 - *y axis will be disease count*
 - *Add a legend to notate which lines are for which disease*
 - *Do a permutation test between the first year and the last year to see if there is statistical significance*
- *Test whether rates differ before and after school starts*
 - *Make new columns for each disease count, splitting it for before and after school starts*
 - *Do a permutation test between before and after school starts to see if there is statistical significance and see where our observed difference lies*
 - *Can also do a bootstrap to find the confidence interval and see if a difference of 0 is observed*
- *Compare high- versus low-vaccination years*
 - *Do a box plot for each disease reporting disease count across high- versus low-vaccination years*
 - *Do a permutation test between high- and low-vaccination years to see if there is statistical significance*